Large and Fast: Exploiting Memory Hierarchy: Part I

CSE 4305: Computer Organization and Architecture

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Preliminary Discussion of the Logical Design of an Electronic Computing Instrument, 1946

A. W. Burks, H. H. Goldstine, and J. von Neumann

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- Spatial locality
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Taking Advantage of Locality

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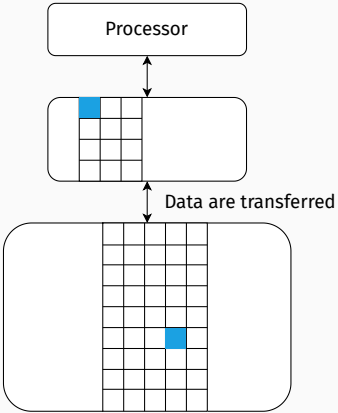
- Copy recently accessed (and nearby) items from disk to smaller DRAM memory (main memory)
- Copy more recently accessed (and nearby) items from DRAM to smaller SRAM memory (cache memory attached to the CPU)

Basic Structure of a Memory Hierarchy

Speed		Size	Cost($/bit)	Current Technology
	Processor			
Fastest	Memory	Smallest	Highest	SRAM
↑	Memory	↓	↑	DRAM
Slowest	Memory	Biggest	Lowest	Magnetic disk or Flash memory

Memory Hierarchy Levels and Terminologies



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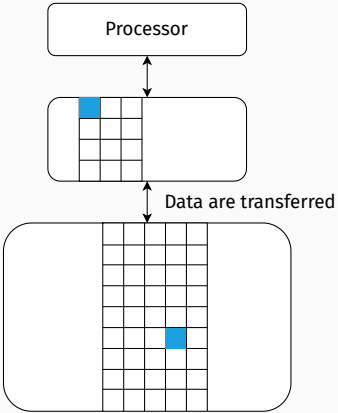

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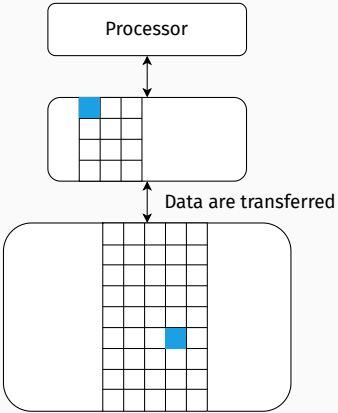

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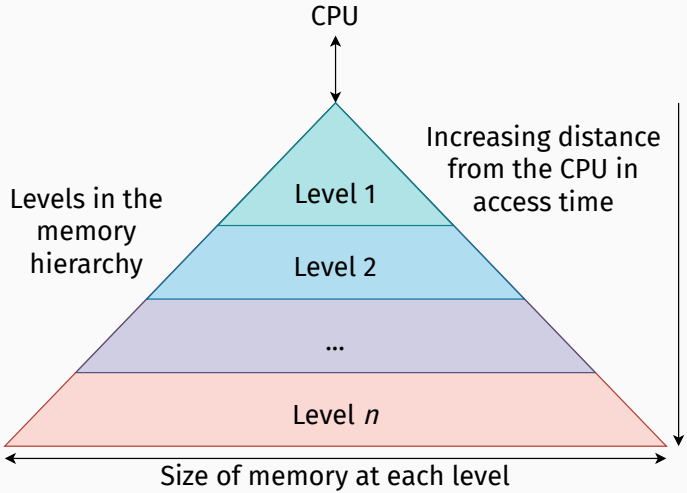
- If accessed data is absent —
  - Miss: block copied from lower level
    - Time taken: miss penalty
    - Miss ratio: $\frac{misses}{accesses} = 1 - hit\ ratio$
  - Then accessed data supplied to the upper-level

Memory Hierarchy Levels Structure



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Ideal Memory

- Access time of SRAM
- Capacity and cost/GB of disk

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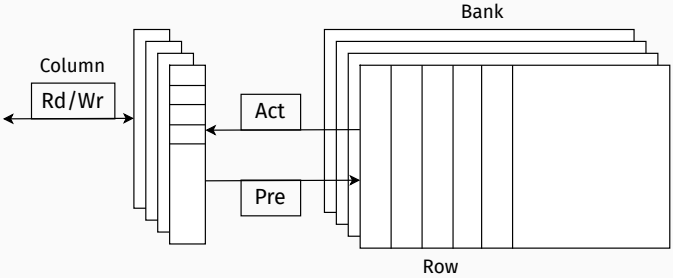
- Consumes very little power in standby
- Once used as separate cache chips, SRAM caches are now integrated on the processor due to Moore's Law

DRAM Technology

Data stored as a charge in a capacitor



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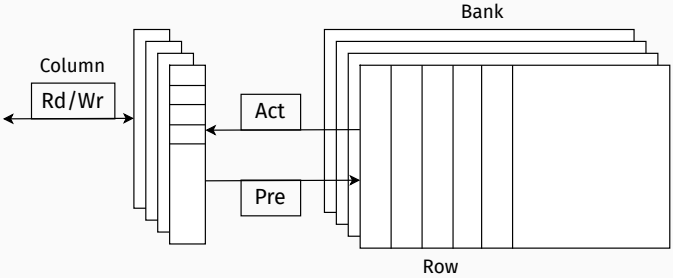
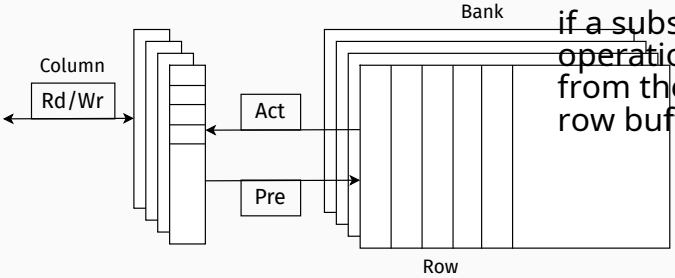


DRAM Technology

Data stored as a charge in a capacitor

- Single transistor used to access the charge
- Must periodically be refreshed
  - Read contents and write back
  - Performed on a DRAM row

* every dram bank contains one row buffer
* when cpu asks for data in a specific cell, full row gets activated
* full row is copied into the row buffer
* then the data in the required cell is returned



if a subsequent operation wants data from the same row -> row buffer hit

Advanced DRAM Organization I

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  - Eliminates the time for the memory and processor to synchronize
- Double Data Rate (DDR) SDRAM
  - Transfer on both rising and falling clock edges — twice as much bandwidth
  - A DDR4-3200 DRAM can do 3,200 million transfers per second, which means it has a 1600-MHz clock

Advanced DRAM Organization II

- Address Interleaving
  - Internally organized to read or write from multiple banks, each having its own row buffer
  - With four banks, there is just one access time and then accesses rotate between the four banks to supply four times the bandwidth

Sequential memory blocks go across different banks so that multiple banks can operate in parallel since each bank has only one row buffer

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- Dual Inline Memory Module (DIMM)
  - Memory for servers, commonly sold on small boards
  - Typically contain 4–16 DRAMs, organized to be 8 bytes wide for server systems

DRAM Performance Factors

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- DRAM Banking
  - Allows simultaneous access to multiple DRAMs
  - Improves bandwidth

DRAM Generations

Year Introduced	Chip Size	$ per GiB	Total Access Time to a New Row/Column	Average Column Access Time to Existing Row
1980	64 Kibibit	$6,480,000	250 ns	150 ns
1983	256 Kibibit	$1,980,000	185 ns	100 ns
1985	1 Mebibit	$720,000	135 ns	40 ns
1989	4 Mebibit	$128,000	110 ns	40 ns
1992	16 Mebibit	$30,000	90 ns	30 ns
1996	64 Mebibit	$9,000	60 ns	12 ns
1998	128 Mebibit	$900	60 ns	10 ns
2000	256 Mebibit	$840	55 ns	7 ns
2004	512 Mebibit	$150	50 ns	5 ns
2007	1 Gibibit	$40	45 ns	1.25 ns
2010	2 Gibibit	$13	40 ns	1 ns
2012	4 Gibibit	$5	35 ns	0.8 ns
2015	8 Gibibit	$7	30 ns	0.6 ns
2018	16 Gibibit	$6	25 ns	0.4 ns

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  - Smaller, lower power, more rugged
  - But more $/GB (between disk and DRAM)

Flash Types

- NOR Flash: Each memory cell connected directly to bit lines (like NOR gates)
  - Very fast read access
  - Random byte-level access
  - Slower erase and write
  - Larger cell size → lower density
  - More expensive per bit
  - Commonly used in BIOS/UEFI/microcontroller firmware, boot ROMs, etc.

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- NAND Flash: Cells connected in series (like NAND gates)
  - Data is accessed in pages and erased in blocks, no true random byte access
  - Very high storage density
  - Much cheaper per bit
  - Fast sequential read/write but slower random access
  - Requires error correction (ECC)
  - Commonly used in SSDs, USB flash drives, SD cards, eMMC/UFS storage, etc.

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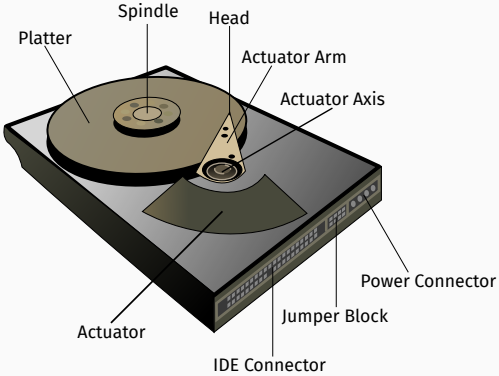
- Such controllers can also improve yield by mapping out incorrectly manufactured memory cells

Disk Storage

Non-volatile, rotating magnetic storage



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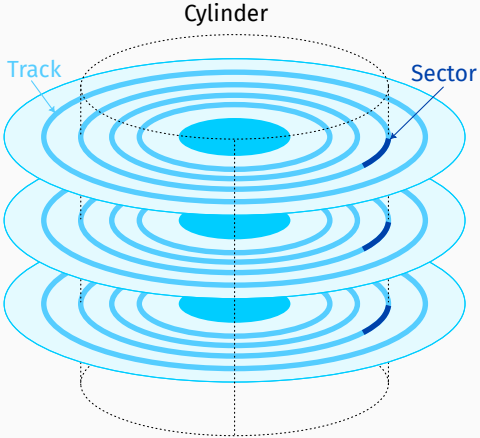

Physical Structure and Data Organization

- Physical Structure
  - Consists of collection of platters rotating on a spindle
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- Head Movement and Cylinders
  - Heads for each surface connected to move in conjunction
  - Cylinder refers to all tracks under heads at a given point

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- Stage 3: Transfer Time
  - Time to transfer a block of bits (150 to 250 MB/sec in 2020)
  - Function of sector size, rotation speed, and recording density
  - Built-in controller cache allows higher transfer rates

Disk Access Example

- Calculate the average read time for a disk access given the following performance parameters:
  - Data size: 512 bytes
  - Rotation speed: 15,000 RPM
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- If actual average seek time is $1ms$, average read time would be $3.205ms$

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- Slower access than flash ($1000\times$) or DRAM ($100,000\times$)
- Cheaper per bit with very high storage capacity
- Non-volatile like flash but no write wear-out problem

The Basics of Cache

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X_3

Before reference to X_n

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Before reference to X_n	After reference to X_n

- Two questions:
  1. How do we know if a data is present?
  2. How do we find it if it is present?

Direct Mapped Cache

- Location determined by the memory address

Direct Mapped Cache

- Location determined by the memory address
- Direct mapped: only one choice

$$\textit{Block Index} = (\textit{Block Address}) \textit{ modulo } (\textit{Number of blocks in the cache})$$

Direct Mapped Cache

```
N % 2^k = k lowest bits
of N
10110 % 2^3 = 110
```

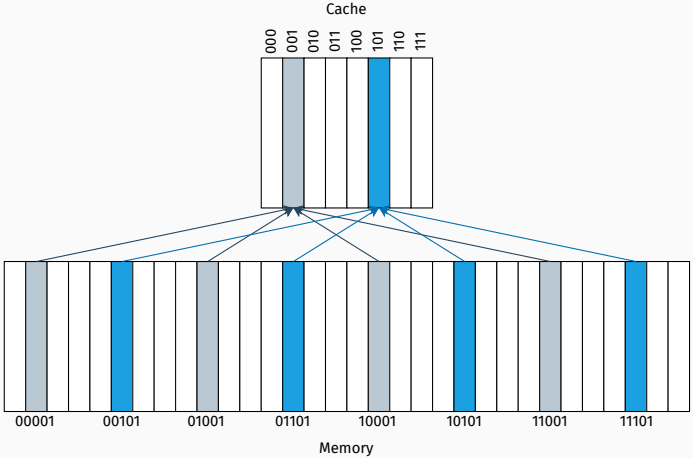
- Location determined by the memory address
- Direct mapped: only one choice

  Block Index = (Block Address) modulo (Number of blocks in the cache)

- If the number of blocks is a power of two, we can simply use lower-order address bits

Direct mapping determines which cache line a retrieved memory block is placed into using a modulo ops. if the number of blocks in the cache is the k-th power of 2 then the result of x 2^k is simply the lowest k bits so it doesnt strain the hardware

Direct Mapped Cache Example



A direct-mapped cache with eight entries showing memory words between 0 and 31 that map to the same blocks

Tags and Valid Bits

- How do we know which particular block is stored in a cache location?

Tags and Valid Bits

```
1101 |010 | 1000
Physical Adress = tag + idx
1101010
Cache Line also has a valid bit which
tells that the data is not garbage
```

- How do we know which particular block is stored in a cache location?
  - Store block address as well as data
  - Only the higher-order bits are necessary (non-redundant)
  - Called the tag

The CPU generates a memory address which the cache interprets as | Tag | Index | Offset |.
The index selects the cache line containing a block of words, and the offset selects the specific word inside that block. The tag contains the upper bits of the memory address that identify the memory block. The cache data is valid if the tag from the CPU address matches the stored tag in the cache line and the valid bit is set.

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  - Called the tag
- What if there is no valid data in a location?
  - Use valid bit — 1 means present (valid), 0 means absent (invalid)
  - Initially set to 0

Cache Example

Cache

Index	V	Tag	Data
000	N		
001	N		
010	N		
011	N		
100	N		
101	N		
110	N		
111	N		

Look-ups

Dec Word Addr	Binary Addr	Hit/Miss	Cache Block

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3	00 011	Miss	011
16	10 000	Hit	000

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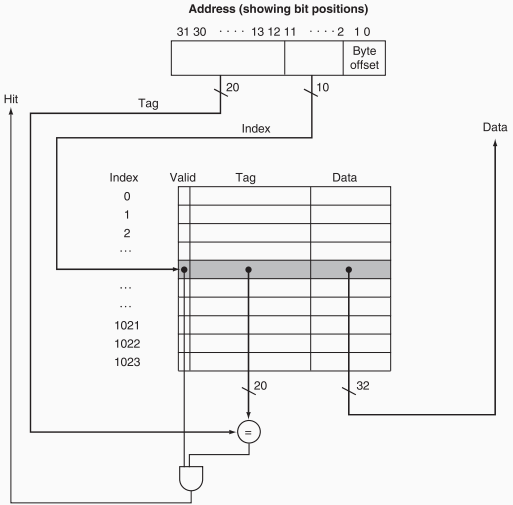
16	10 000	Hit	000
18	10 010	Miss	010

Address Subdivision

valid | tag | data



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we are accessing memory at byte level not word level so we also need 2 extra bits to specify which byte to select from the selected word in the selected cache line

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  $$2^n \times (2^m \times 32 + (32 - n - m - 2) + 1) = 2^n \times (2^m \times 32 + 31 - n - m)$$
- Naming conventions specify cache size based only on the data portion

```
4 words per block
2^2 words ->
00 -> word1
01 -> word2
10 -> word3
11 -> word4

00 -> byte1
01 -> byte2
10 -> byte3
11 -> byte4
```

Example: Bits in a Cache

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- Complete cache size is $2^{10} \times (128 + 50 + 1) = 179$ Kibibits
- Total physical size is roughly 1.4 times the required data storage

```
1 kibibit = 1024
bit (2^10 bits)
```

```
cache line-
1 + 50 + 128
179 / 128 = 1.398
```

Example: Mapping an Address

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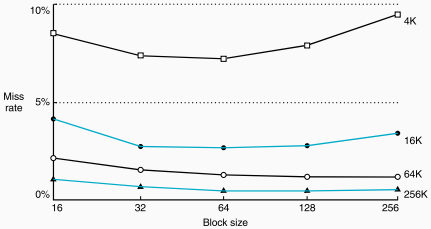
- Cache block maps to 75 **mod** 64 = 11
- This specific block holds all addresses between 1200 and 1215

word1 1200 - 1203
word2 1204 - 1207
word3 1208 - 1211
word4 1212 - 1215

Block Size Trade-offs

- Larger blocks lower miss rates by exploiting spatial locality



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(each line represents a different sized cache)

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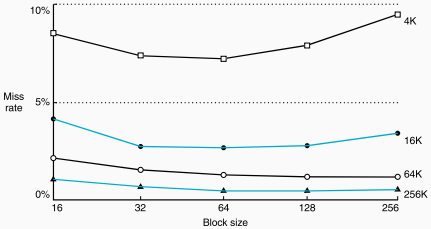

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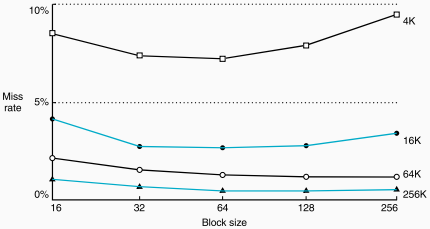

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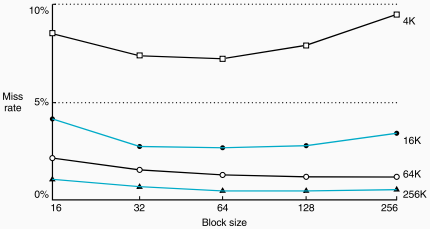

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Block Size Trade-offs

```
cache can hold N bits
if i increase block size i can hold less
blocks
```

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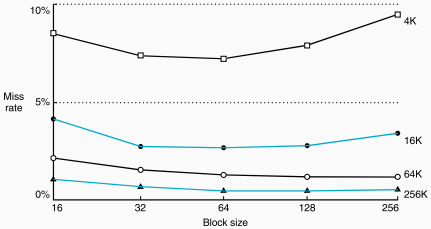
- Performance degrades when the miss penalty increase overwhelms the miss rate decrease



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- Separate controller initiates memory access and refills cache

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- Early restart resumes execution right after the requested word is returned
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Hiding Transfer Time

```
cpu gives the required mem address and asks for
word2
early restart - word1 - word2 (cpu starts) - word3
- word4
CWF + Early Restart - word2 (cpu starts) - word3 -
word4 - word1
```

- Transfer times can be partially hidden to reduce the effective miss penalty
- Early restart resumes execution right after the requested word is returned
  - Works exceptionally well for sequential instruction accesses
  - Less effective for unpredictable data accesses
- Critical word first transfers the requested word before the rest of the block
  - Remaining words are transferred sequentially and wrap around to the beginning
  - Slightly faster than early restart but faces identical limitations

early restart- cpu starts working immediately after the requested word is retrieved, doest wait for whole block
critical word first- requested block is transferred first

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- Data miss handling is essentially identical

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Handling Writes: Consistency

```
cpu requests memA
memA gets written to the cache
cpu may change the value of memA
the value of new memA is written in cache
need to ensure to update the original value of
memA in the RAM
```

- Writing only to cache makes it inconsistent with memory
- Write-through: Update both cache and main memory simultaneously
- Write miss procedure in write-through:
  - Fetch block from memory
  - Overwrite specific word in cache
  - Write word to main memory

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- Processor stalls only if write buffer is full
- Multiple entries in buffer help handle write bursts

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- Modified block written to memory only when replaced
- Improves performance if write generation exceeds memory speed
- More complex to implement than write-through

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- No write allocate: update memory directly without caching
- No write allocate useful when writing entire blocks (e.g., OS zeroing pages)
- Policy can sometimes be changed on per-page basis

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Implementing Write-Back

```
cache line - line1 *line2* line3 line4
taking line2 in a store buffer, the line2 gets evicted by
the new memory. Then buffer updates the value for the old
line2 in the RAM.
```

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- Implementation options:
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- Write-back buffer holds modified blocks during replacement to reduce penalty

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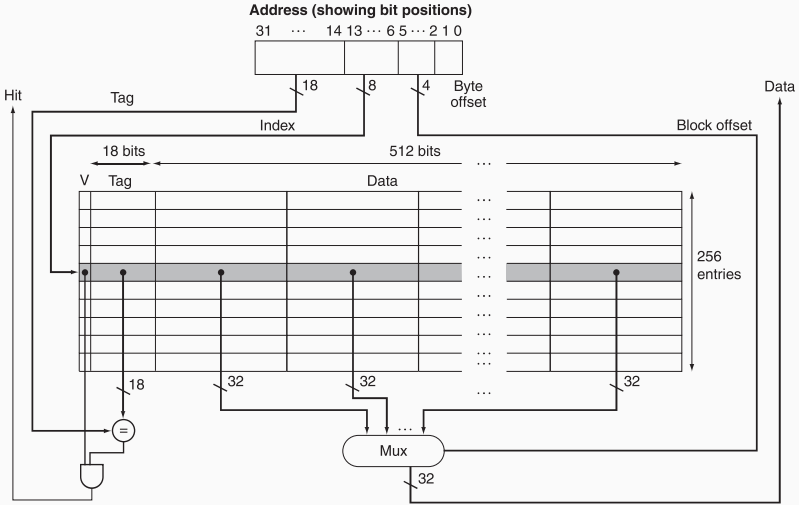
  - Each 16KiB — 256 blocks $\times$ 16 words/block
  - D-cache offers both write-through and write-back, OS gets to decide the strategy
- SPEC2000 miss rate:
  - I-cache: 0.4%
  - D-cache: 11.4%
  - Effective combined (weighted average): 3.2%

Example: Intrinsity FastMATH Cache Organization



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- Bandwidth improvements needed for efficiency —
  - Wider memory and interleaving
  - Optimized DRAM interfaces for burst mode

```
in ssd, data is stored in pages
each page has a data bank
data is first transferred to data bank,
then transferred to RAM
data is spread across multiple banks to
reduce the load for each bank. So each
bank acts in parallel, increasing
throughput.
```

Measuring and Improving Cache Performance

Measuring Cache Performance

- CPU time can be divided into two components —
  - CPU execution clock cycles: Normally includes cache hit time
  - Memory-stall clock cycles: Mainly from cache misses

$$\boxed{\textit{CPU time} = (\textit{CPU execution cycles} + \textit{Memory-stall cycles}) \times \textit{Clock cycle time}}$$

read miss - cache miss (data required not in cache)
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- With simplified assumptions, we can define memory-stall clock cycles as —

$$\boxed{\textit{Memory-stall cycles} = (\textit{Read-stall cycles} + \textit{Write-stall cycles})}$$

Where,

- $\textit{Read-stall cycles} = \frac{\textit{Reads}}{\textit{Program}} \times \textit{Read miss rate} \times \textit{Read miss penalty}$
- $\textit{Write-stall cycles} = \left(\frac{\textit{Writes}}{\textit{Program}} \times \textit{Write miss rate} \times \textit{Write miss penalty}\right) + \textit{Write buffer stalls}$

Memory-stall Clock Cycles

- Considering write buffer stalls are negligible, in most write-through organizations, the read and write miss penalties are same.

$$\begin{aligned}\therefore \textit{Memory-stall cycles} &= \frac{\textit{Memory access}}{\textit{Program}} \times \textit{Miss rate} \times \textit{Miss penalty} \\ &= \frac{\textit{Instructions}}{\textit{Program}} \times \frac{\textit{Misses}}{\textit{Instruction}} \times \textit{Miss penalty}\end{aligned}$$

Cache Performance Example

- Given,
  - I-cache miss rate = 2%
  - D-cache miss rate = 4%
  - Miss penalty is = 100 cycles
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- Solution:

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  - Base CPI without stalls (ideal cache) = 2
  - Load and stores are 36% of instructions
- Solution:
  - Memory miss cycles count in terms of I instructions:
    - I-cache: $I \times 2\% \times 100 = 2.00I$
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  - Total CPI including memory stalls $= 2 + 3.44 = 5.44$
  - Perfect cache/Ideal CPU is $^{5.44}/_{2} = 2.72$ times faster
  - If base CPI was 1, a perfect system cache would be $^{1+3.44}/_{1} = 4.44$ times faster

```
CPI(including stalls) =  CPI(without stalls) +
(No.of Stalled cyles / Instructions)

=> CPI(including stalls) = 2 + (3.44I/I)
=>CPI(including stalls) = 5.44

As the CPI of the CPU increases, the miss penalty
remains constant. Since the clock speed of the CPU
is greater, the number of stalls the CPU has to
undergo is greater.
```

Average Access Time

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Example

Find the AMAT for a processor with a 1 ns clock cycle time, a miss penalty of 20 clock cycles, a miss rate of 5%, and a cache access time (including hit detection) of 1 clock cycle. Assume that the read and write miss penalties are the same and ignore other write stalls.

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Solution:

$$AMAT = 1 + 0.05 \times 20 = 2ns\ (2 \text{ clock cycles})$$

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Associative Caches

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  - Allows a given block to go in any cache entry
  - Requires all entries to be searched at once
  - Comparator associated with each cache entry for parallel search (expensive)

Associative Caches

Schemes other than direct mapping exist for block placement in a cache

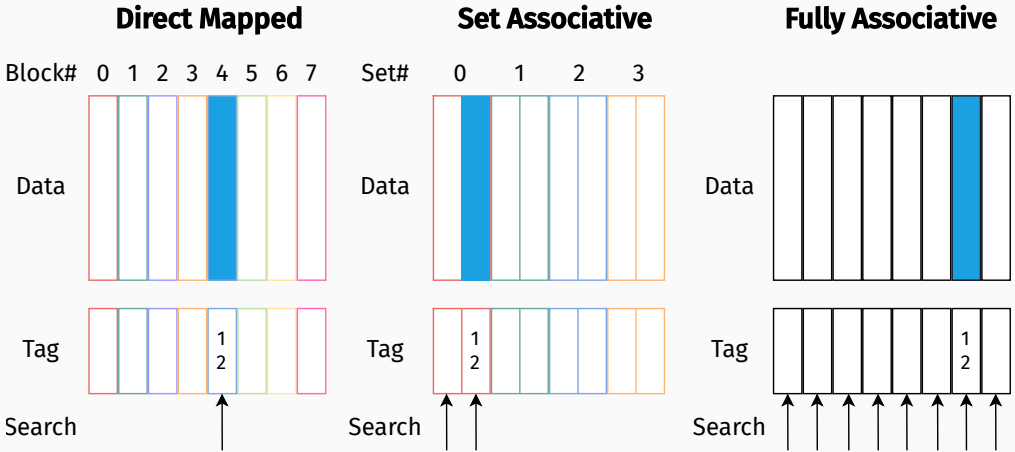
```
n-way where n=3
memA can go to LocationA(blockA1-blockA3)
```

- Fully associative
  - Allows a given block to go in any cache entry
  - Requires all entries to be searched at once
  - Comparator associated with each cache entry for parallel search (expensive)
- *n*-way set associative
  - Each block can be placed within a fixed number of locations (in a set)
  - Each set contains *n* entries
  - Block number and set count determines which set to put in

$$\textit{Block number} \mod \textit{Number of sets in the cache}$$

  - Search all the entries in a given set at once
  - *n* comparators (comparatively less expensive)

Associate Cache Example



Spectrum of Associativity

One-way set associative (direct mapped)

Block	Tag	Data
0		
1		
2		
3		
4		
5		
6		
7		

Two-way set associative

Block	Tag	Data	Tag	Data
0				
1				
2				
3				

Four-way set associative

Block	Tag	Data	Tag	Data	Tag	Data	Tag	Data
0								
1								

Eight-way set associative (fully associative)

Tag	Data	Tag	Data	Tag	Data	Tag	Data	Tag	Data	Tag	Data	Tag	Data	Tag	Data

Associativity Example I

- Compare three small caches consisting of four one-word blocks —
  - Direct mapped, 2-way set associative, and fully associative
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- Direct Mapped ($0 \mod 4 = 0$; $6 \mod 4 = 2$; $8 \mod 4 = 0$)

Address of memory block addressed	Hit or miss	Content of the cache blocks after reference			
		0	1	2	3
0	miss	Memory[0]			
8	miss	Memory[8]			
0	miss	Memory[0]			
6	miss	Memory[0]		Memory[6]	
8	miss	Memory[8]		Memory[6]	

Associativity Example II

- 2-way Set Associative ($0 \mod 2 = 0$; $6 \mod 2 = 0$; $8 \mod 2 = 0$)

Address of memory block addressed	Hit or miss	Content of the cache blocks after reference			
		Set 0	Set 0	Set 1	Set 1
0	miss	Memory[0]			
8	miss	Memory[0]	Memory[8]		
0	hit	Memory[0]	Memory[8]		
6	miss	Memory[0]	Memory[6]		
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Associativity Example III

- Fully Associative

Address of memory block addressed	Hit or miss	Content of the cache blocks after reference			
		Block 0	Block 1	Block 2	Block 3
0	miss	Memory[0]			
8	miss	Memory[0]	Memory[8]		
0	hit	Memory[0]	Memory[8]		
6	miss	Memory[0]	Memory[8]	Memory[6]	
8	hit	Memory[0]	Memory[8]	Memory[6]	

How Much Associativity?

- Increased associativity decreases miss rate, but with diminishing returns

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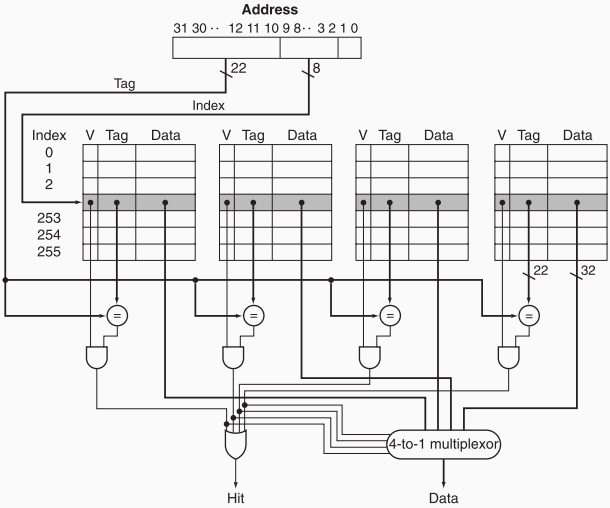
- Increased associativity decreases miss rate, but with diminishing returns
- Simulation of a system with 64KiB D-cache, 16-word blocks (Intrinsity FastMATH) in SPEC2000 —

Associativity	Data Miss Rate
1	10.3%
2	8.6%
4	8.3%
8	8.1%

Set Associative Cache Organization



Replacement Policy

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  - Simple for 2-way, manageable for 4-way, too complex/costly beyond that
- Random
  - Gives approximately the same performance as LRU for high level of associativity

Multilevel Caches

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- Primary cache is attached to the CPU — small, but fast
- L2 cache services misses from the primary cache — larger, slower, but still faster than the main memory
- Main memory services L2 cache misses
- Most modern high-end systems (generally multi-core) include an L3 cache as well
  - L1 is private to a processor core, separate I- and D-caches
  - L2 also usually private to a processor core, generally stores both instruction and data
  - L3 is shared across the different cores

Multilevel Cache Example I

- Given,
  - CPU base CPI $= 1$, clock-rate $= 4GHz$
  - Miss rate/instruction $= 2\%$
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  - Main memory access time $= 100ns$
- Considering just the primary cache,
  - Miss penalty $= {}^{100ns}/_{0.25ns} = 400$ cycles (access time converted to cycles using clock-rate)
  - Effective CPI $= 1 + 2\% \times 400 = 9$

Multilevel Cache Example II

- Now consider an added L2 cache with,
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$$= 1 + \text{Primary stalls per instruction} + \text{Secondary stalls per instruction}$$

$$= 1 + 2\% \times 20 + 0.5\% \times 400 = 3.4$$

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  $= 1 + \text{Primary stalls per instruction} + \text{Secondary stalls per instruction}$

  $= 1 + 2\% \times 20 + 0.5\% \times 400 = 3.4$
- Performance ratio between without/with secondary cache $= {}^{9}/_{3.4} = 2.6$

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  - L1 block size becomes smaller than L2 block size
- Considering L1 cache filters access, the local miss rate (miss in a specific cache level) of L2 cache can be quite high compared to the global miss rate (not found in any cache)

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- Effect of miss depends on the program data flow — much harder to analyze and requires system simulations

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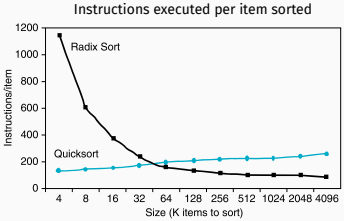
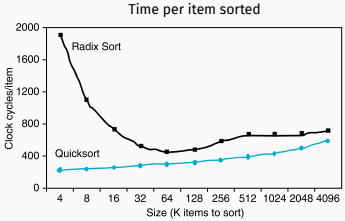
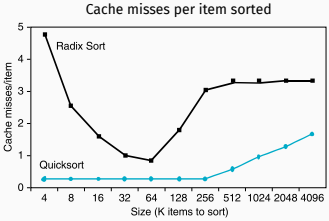






Software Optimization via Blocking

- Memory hierarchy is crucial for program performance
- Software optimizations can drastically improve execution speed
- The main goal is to reuse data already in the cache
- This lowers miss rates by improving temporal locality

The Array Access Problem

```
a b c d e f
g h i j k l
m n o p q r
```

- Sequential memory access yields the best performance
- But some multi-dimensional arrays, like in the case of matrix multiplication, are accessed by rows and others by columns
- Storing arrays in row-major or column-major order does not fix this when both rows and columns are used in every loop iteration

What is Blocking?

```
A x B
a1  a2  a3  a4
a5  a6  a7  a8
a9  a10 a11 a12
a13 a14 a15 a16

A =
Ba1 Ba2
Ba3 Ba4

B =
Bb1 Bb2
Bb3 Bb4
```

- Instead of full rows or columns, we use submatrices or blocks
- Blocked algorithms operate purely on these smaller sections
- Maximizes accesses to data while it is loaded in the cache
- Keeps data active before it gets replaced by new memory loads

Standard Matrix Multiplication

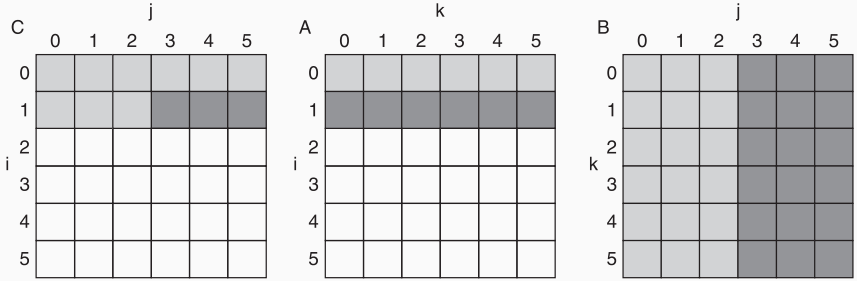
- Consider the multiplication of three $N \times N$ matrices ($C = A \times B$)
- The algorithm reads all elements of matrix B
- It reads the same elements in one row of matrix A repeatedly
- It writes to one row of matrix C

Standard Matrix Multiplication Code (Unblocked)

Inner loops for standard matrix multiplication —

```
for (int i = 0; i < n; ++i) {
    for (int j = 0; j < n; ++j) { // Inner loop from here
        double cij = C[i+j*n]; /* cij = C[i][j] */
        for (int k = 0; k < n; k++ )
            cij += A[i+k*n] * B[k+j*n];
        C[i+j*n] = cij; /* C[i][j] = cij */
    }
}
```

Access Pattern in the Unblocked Code



A snapshot of the three arrays (matrices) C, A, and B when $N = 6$ and $i = 1$; the age of accesses to the array elements is indicated by shade: white means not yet touched, light means older accesses, and dark means newer accesses.

Cache Misses in Unblocked Code

- Number of capacity misses depends on N and cache size
- If the cache holds all three matrices, performance is optimal
- If the cache is too small, misses occur heavily for B and C
- Worst case requires $2N^3 + N^2$ memory words accessed

The Blocking Solution

- We change the code to compute on a submatrix
- The submatrix size is defined by a blocking factor
- The blocking factor is called BLOCKSIZE
- This ensures the accessed elements easily fit inside the cache

Blocked Matrix Multiplication Code

```
#define BLOCKSIZE 32

void do_block (int n, int si, int sj, int sk,  const double
    *A, const double *B, double *C) {
     for (int i = si; i < si+BLOCKSIZE; ++i)
         for (int j = sj; j < sj+BLOCKSIZE; ++j) {
             double cij = C[i+j*n];
             for(int k = sk; k < sk+BLOCKSIZE; k++)
                 cij += A[i+k*n] * B[k+j*n];
             C[i+j*n] = cij;
         }
}
```

Invoking the Blocked Code

The main function calls the block function repeatedly —

```
void dgemm (int n, const double* A, const double* B, double*
    C) {
    for (int sj = 0; sj < n; sj += BLOCKSIZE)
        for (int si = 0; si < n; si += BLOCKSIZE)
            for (int sk = 0; sk < n; sk += BLOCKSIZE)
                do_block(n, si, sj, sk, A, B, C);
}
```

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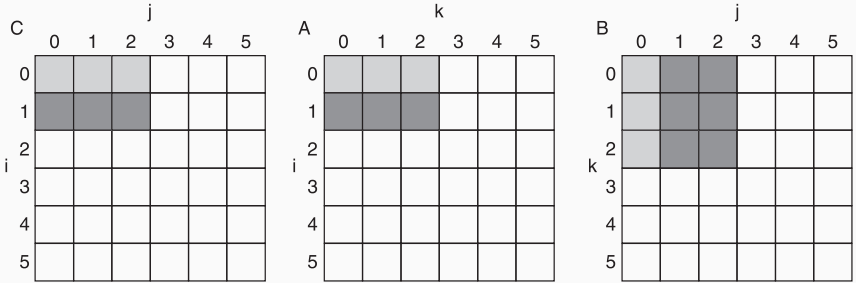

The age of accesses to the arrays C, A, and B when BLOCKSIZE $= 3$; in contrast to the unblocked version, fewer elements are accessed

Performance Improvements

- Total memory accessed becomes $2N^3/\text{BLOCKSIZE} + N^2$
- Performance improves roughly by a factor of BLOCKSIZE
- Matrix A benefits heavily from spatial locality
- Matrix B benefits heavily from temporal locality
- Small block sizes can also improve register allocation

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- Software Optimization: Algorithms can be restructured via blocking to improve temporal locality and cache effectiveness

Dependable Memory Hierarchy

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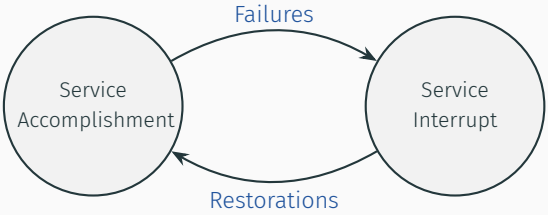

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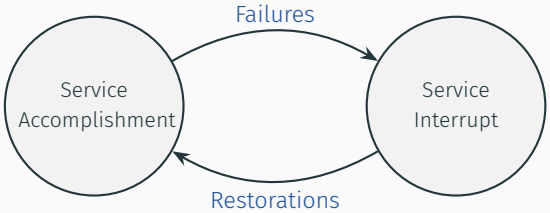
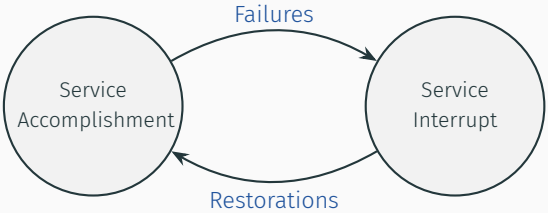


- Failures can be permanent or intermittent (harder to deal with)

Dependability and Failure

- Core idea for dependability is redundancy
- Systems alternate between two states —
  - Service accomplishment: service delivered as specified
  - Service interruption: delivered service differs from specification
- Transitions between states —



- Failures can be permanent or intermittent (harder to deal with)
- Two associated terms, reliability and availability

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- Large MTTF values can be misleading
- AFR provides better intuition for large-scale systems

Example: MTTF vs AFR

- Assume disk MTTF is 1,000,000 hours (114 years)
- Assume warehouse-scale computer with 50,000 servers (100,000 disks)
- 1 year equals 8760 hours
- AFR is $^{8760}/_{1000000} = 0.876\%$
- Expected failures equal $100,000 \times 0.00876 = 876$ disks per year
- This averages to more than two disk failures per day

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- Availability is measured in nines (for example 5 nines is 99.999%, about 5 minutes of repair per year)

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- Three ways to improve MTTF —
  1. Fault avoidance: prevent occurrence by construction
  2. Fault tolerance: use redundancy to allow service despite faults
  3. Fault forecasting: predict and replace before failure

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- Example: distance between 011011 and 001111 is two
- A minimum distance of two allows detection of single-bit errors
- Parity code counts the number of 1s to detect errors
  - Adds a parity bit to make the total number of 1s even (even parity)
  - Mismatch between stored parity and calculated parity signals an error

Example: Simple Parity

- Data: 31_{ten} or 00011111_{two} (five 1s)
- To achieve even parity, add a 1 as the parity bit: 000111111_{two}
- Single bit inversion (e.g., MSB flips)
  - Read as 100111111_{two} (seven 1s)
  - Odd parity calculated, but even parity expected: error detected
- Double bit inversion
  - Read as 110111111_{two} (eight 1s)
  - Even parity calculated: error not detected
- 1-bit parity cannot correct errors, only detect them

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- Data bits occupy all other positions (3, 5, 6, 7, 9, 10, 11, 12...)
- Parity bit positions determine the sequence of data bits they check —
  - Bit 1 checks positions where rightmost address bit is 1 (1, 3, 5, 7, 9, ...)
  - Bit 2 checks positions where second address bit is 1 (2, 3, 6, 7, 10, ...)
  - Bit 4 checks positions where third address bit is 1 (4–7, 12–15, ...)
  - Bit 8 checks positions where fourth address bit is 1 (8–15, 24–31, ...)

Hamming ECC Coverage Table

Bit Position		1	2	3	4	5	6	7	8	9	10	11	12
Encoded Data Bits		p1	p2	d1	p4	d2	d3	d4	p8	d5	d6	d7	d8
Parity Bit Coverage	p1	X		X		X		X		X		X	
	p2		X	X			X	X			X	X	
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- If results are 0000, there is no error
- If results form a non-zero binary number (e.g., $1010_{two} = 10_{ten}$), that number identifies the exact bit position of the error
- Correct the error by inverting that specific bit

Example: Hamming ECC Calculation

- Data byte: 10011010_{two}
- 12-bit pattern layout (p = parity, d = data): $p_1\ p_2\ d_1\ p_4\ d_2\ d_3\ d_4\ p_8\ d_5\ d_6\ d_7\ d_8$
- Initial state: $p_1\ p_2\ 1\ p_4\ 0\ 0\ 1\ p_8\ 1\ 0\ 1\ 0$
- Calculate parity bits for even parity
  - Position 1 (1,3,5,7,9,11): $\underline{p_1}\ p_2\ \underline{1}\ p_4\ \underline{0}\ 0\ \underline{1}\ p_8\ \underline{1}\ 0\ \underline{1}\ 0 \rightarrow$ four 1s $\rightarrow p_1 = 0$
  - Position 2 (2,3,6,7,10,11): $0\ \underline{p_2}\ \underline{1}\ p_4\ 0\ \underline{0}\ \underline{1}\ p_8\ 1\ \underline{0}\ \underline{1}\ 0 \rightarrow$ three 1s $\rightarrow p_2 = 1$
  - Position 4 (4,5,6,7,12): $0\ 1\ 1\ \underline{p_4}\ \underline{0}\ \underline{0}\ \underline{1}\ p_8\ 1\ 0\ 1\ \underline{0} \rightarrow$ one 1 $\rightarrow p_4 = 1$
  - Position 8 (8,9,10,11,12): $0\ 1\ 1\ 1\ 0\ 0\ 1\ \underline{p_8}\ \underline{1}\ \underline{0}\ \underline{1}\ \underline{0} \rightarrow$ two 1s $\rightarrow p_8 = 0$
- Final encoded word: 011100101010

Example: Correcting an Error

- Original word: 011100101010
- Assume bit 10 inverts during transmission: 011100101110
- Recalculate parities on received word
  - Parity 1 group: $\underline{0}$ 1 $\underline{1}$ 1 $\underline{0}$ 0 $\underline{1}$ 0 $\underline{1}$ $\underline{1}$ $\underline{1}$ 0 (four 1s) $\rightarrow$ even (OK)
  - Parity 2 group: 0 $\underline{1}$ $\underline{1}$ 1 0 $\underline{0}$ $\underline{1}$ 0 1 $\underline{1}$ $\underline{1}$ 0 (five 1s) $\rightarrow$ odd (error)
  - Parity 4 group: 0 1 1 $\underline{1}$ $\underline{0}$ $\underline{0}$ $\underline{1}$ 0 1 1 1 $\underline{0}$ (two 1s) $\rightarrow$ even (OK)
  - Parity 8 group: 0 1 1 1 0 0 1 $\underline{0}$ $\underline{1}$ $\underline{1}$ $\underline{1}$ $\underline{0}$ (three 1s) $\rightarrow$ odd (error)
- Parity bits 2 and 8 are incorrect
- $2 + 8 = 10$, so bit 10 is wrong
- Correct by inverting bit 10 back to 0

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- Standard in modern server memory (e.g., 72-bit DIMMs for 64-bit data)

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- Chipkill (IBM) or SDDC (Intel)
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  - Reduces unrecoverable failures drastically compared to standard SEC/DED
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- Network Error Handling
  - Cyclic Redundancy Check (CRC) handles bursts of errors using binary division
  - Reed-Solomon codes use Galois fields and polynomials for complex multi-bit transmission errors

To be continued in *Large and Fast: Exploiting Memory Hierarchy: Part II*!